

Microeconomics

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Asymmetric information-Adverse Selection

Adverse Selection

- A firm hiring a worker may know less than the worker does about the worker's ability and hence about her productivity
- The buyer of a used car may know less than the seller does about the quality of the car
- A bank financing an entrepreneur's project may know less than the entrepreneur does about the entrepreneur's skills and hence about the profitability of the project
- A company providing car insurance may know less than the driver does about the driver's driving skills

Adverse Selection

- **Adverse selection** is the phenomenon under which the informed side of a transaction has information that the other side does not have about some aspect of product quality and this leads to a poor performance of the market.
- Not possible to separate low quality from high quality products: only one price
- This leads to:
 - the disappearance of high quality products from the market
 - A higher price for low quality products than under symmetric information
 - ➔ inefficiency

Adverse Selection

- The basic reason for inefficiency: we have several goods but only one price
- In the examples above, all goods look the same before they are bought
- This prevents the creation of two separate markets, each with its own price
- No single price can simultaneously persuade higher-quality sellers to trade at that price and dissuade the lower-quality ones from doing so

Example: Akerlof's market for "lemons"

- Suppose that, in the market for used cars, there are two types of cars on sale: "good" cars G and "bad" cars B (also known as "lemons"); suppose also that there are N cars to sell ($N/2$ are good and $N/2$ are bad), and, for simplicity, infinitely many potential buyers
- Assume the owners are **willing to sell** the good and bad cars respectively at:

$$P_G^s = 4$$

$$P_B^s = 2$$

Example: Akerlof's market for "lemons"

- The customers are **willing to buy** the good and bad cars respectively at:

$$p_G^d = 4.4$$

$$p_B^d = 2.2$$

- Customers cannot see the quality of the used car before buying it
- Both the buyers and the sellers **know** that the **probability** of buying a lemon is $1/2$, i.e. that there are $N/2$ good cars and $N/2$ bad cars on the market

Full Information Equilibrium

- If the quality of a used car was not a hidden characteristic, in the full-information equilibrium, **both types** of cars would be sold at some equilibrium price. What are the relevant conditions that have to be satisfied?
 - For the bad car?
 - $P \in [P^s, P^b] = [2, 2.2]$
 - For the good car?
 - $P \in [P^s, P^b] [4, 4.4]$
- Hence, in a full-information equilibrium **all** of the N cars would be traded, and the economy would reach a **Pareto-efficient allocation**
 - We would have two markets, ones for good cars one for bad
 - Aggregate surplus would be maximum possible

Asymmetric Information Equilibrium

- If the quality of a used car is a hidden characteristic, the owners are likely to know more about the quality of the used car than the buyers; for simplicity, we assume that the owners perfectly observe the cars' quality, while the buyers cannot observe it
- Hence, from the **buyers' perspective**, what is the **expected value, EV**, of a car?
 - $EV = 0.5 \times 2.2 + 0.5 \times 4.4 = 3.3$
- *EV* is the **maximum price** that any customer will ever pay for a used car in an asymmetric-information equilibrium

Asymmetric Information Equilibrium

- Will the good cars ever enter the market?
- $P_G^S = 4 > 3.3 = EV$
- $P_B^S = 2 < 3.3 = EV$
- The reason why good cars never enter the market is that the highest price that the customers are willing to pay is clearly below the minimum price at which the owners are willing to sell them!!

Asymmetric Information Equilibrium

- Given that all relevant pieces of information, apart from the cars' hidden quality, are perfectly known, the buyers realize that all cars offered on the market have to be “bad” → then they will pay at most 2.2 for a car
- Therefore, in an asymmetric-information equilibrium only bad cars will be traded at some equilibrium price:
 - *Lemons drive good cars out of the market*
- This is a case of **Adverse Selection**: in the used-car market, the buyers (the uninformed party) get an adverse selection of the sellers' (the informed party's) good

Asymmetric Information Equilibrium

The asymmetric-information equilibrium is clearly **Pareto-inefficient**: only 50% of the cars is actually traded, even if the buyers and the sellers are willing to trade the remaining 50% at a higher price

Asymmetric Information Equilibrium

- Suppose instead that $P_G^S = 3.1$. Then $EV = 3.3 > 3.1$ and $EV > 2$
- In this case, good and bad cars will enter the market. What about the equilibrium price?
- Here the maximum price that customers are willing to pay is the expected value EV
- Given that the cars' quality is a hidden characteristic, the owners of bad cars will **never** voluntarily reveal that they are selling lemons, and therefore the minimum price at which all owners are willing to sell their cars is p_G^S
- Equilibrium price $P \in [p_G^S, EV] = [3.1, 3.3]$
- All cars are traded in equilibrium, efficiency is reached, but low quality cars are too expensive!

Long-run effects

- But consider now the owners' profit levels: clearly, in equilibrium the owners of bad cars make much higher profits than the unlucky owners of good cars
- If selling bad cars is such a profitable business, we may conclude that even more owners of bad cars will have the incentive to enter in the used-car market in the long run
- The higher incentive to sell their car faced by lemon owners introduces a negative feedback mechanism that leads to a steady decrease in the average quality of a used car, and the steady increase in the probability of buying a lemon
- In the long run, good cars will again completely disappear from the market

Job market

- 2 types of workers
 - H, TR=12000
 - L, TR=6000
- $Q_h^s = 0.05x(W-2000)$
- $Q_l^s = 0.1x(W-2000)$

Observable actions equilibrium

➤ $D^h = S^h$

➤ $Q = 0,05 \times (12000 - 2000) = 500$

➤ $D^l = S^l$

➤ $Q = 0,1 \times (6000 - 2000) = 400$

Asymmetric information equilibrium

- $Q^s = Q_h^s + Q_l^s = 0,15(W - 2000)$
- $ETR = pr(H)12000 + pr(L)6000$
 - $pr(H) = Q_h^s / (Q_h^s + Q_l^s) = 1/3$
 - $ETR = 12000 \times 1/3 + 6000 \times 2/3 = 8000$
- Equilibrium: $Q = 0.15(8000 - 2000) = 900$, of which 300 H & 600 L
- In equilibrium: too many L and few H
- $DWL = (12000 - 8000)100 + (8000 - 6000)100 = 600000$